

Retrieval-Augmented Generation for Document Intelligence: Source-Grounded Question Answering across Research, Enterprise, and Educational Corpora

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Abstract

This paper presents a unified Retrieval-Augmented Generation (RAG) framework for source-grounded question answering across research, enterprise, and educational corpora. The study investigates how chunking strategies, retrieval methods, and generation configurations affect retrieval quality, answer quality, and faithfulness. A common pipeline consisting of preprocessing, chunking, embedding generation, vector retrieval, context construction, and citation-supported answer generation is evaluated across heterogeneous document collections. Illustrative evaluation results suggest that retrieval-augmented approaches outperform sparse retrieval baselines, while semantic chunking and hybrid retrieval provide the strongest performance. The study contributes a cross-domain evaluation framework, separate assessment of retrieval and generation quality, and a domain-sensitive faithfulness analysis.

Keywords

Retrieval-Augmented Generation, Large Language Models, Document Intelligence, Question Answering, Faithfulness

1. Introduction

Large Language Models are increasingly used as interfaces for querying document repositories. However, standalone models may generate unsupported responses. Retrieval-Augmented Generation addresses this limitation by conditioning generation on retrieved evidence. Most prior studies evaluate RAG within a single domain. This work investigates a unified framework across research, enterprise, and educational corpora, focusing on retrieval quality, answer quality, and faithfulness. The study aims to

determine how chunking and retrieval choices affect performance across heterogeneous document collections. Large Language Models are increasingly used as interfaces for querying document repositories. However, standalone models may generate unsupported responses. Retrieval-Augmented Generation addresses this limitation by conditioning generation on retrieved evidence. Most prior studies evaluate RAG within a single domain. This work investigates a unified framework across research, enterprise, and educational corpora, focusing on retrieval quality, answer quality, and faithfulness. The study aims to determine how chunking and retrieval choices affect performance across heterogeneous document collections. Large Language Models are increasingly used as interfaces for querying document repositories. However, standalone models may generate unsupported responses. Retrieval-Augmented Generation addresses this limitation by conditioning generation on retrieved evidence. Most prior studies evaluate RAG within a single domain. This work investigates a unified framework across research, enterprise, and educational corpora, focusing on retrieval quality, answer quality, and faithfulness. The study aims to determine how chunking and retrieval choices affect performance across heterogeneous document collections.

2. Related Work

Prior work explores enterprise knowledge assistants, document intelligence systems, governance-oriented architectures, agentic retrieval systems, and faithfulness evaluation frameworks. Existing studies are largely domain-specific and do not provide a controlled comparison across research, enterprise, and educational corpora. This motivates the need for a unified evaluation framework. Prior work explores enterprise knowledge assistants, document intelligence systems, governance-oriented architectures, agentic retrieval systems, and faithfulness evaluation frameworks. Existing studies are largely domain-specific and do not provide a controlled comparison across research, enterprise, and educational corpora. This motivates the need for a unified evaluation framework. Prior work explores enterprise knowledge assistants, document intelligence systems, governance-oriented architectures, agentic retrieval systems, and faithfulness evaluation frameworks. Existing studies are largely domain-specific and do not provide a controlled comparison across research, enterprise, and educational corpora. This motivates the need for a unified evaluation framework. Prior work explores enterprise knowledge assistants, document intelligence systems, governance-oriented architectures, agentic retrieval systems, and faithfulness evaluation frameworks. Existing studies are largely domain-specific and do not provide a controlled comparison across research, enterprise, and educational corpora. This motivates the need for a unified evaluation framework.

3. Methodology

The proposed framework consists of document preprocessing, chunking, embedding generation, vector indexing, retrieval, context construction, and citation-supported answer generation. Fixed-size, overlapping, and semantic chunking strategies are considered. Retrieval quality and generation quality are evaluated separately to distinguish retrieval failures from generation failures. The proposed framework consists of document preprocessing, chunking, embedding generation, vector indexing, retrieval, context construction, and citation-supported answer generation. Fixed-size, overlapping, and semantic chunking strategies are considered. Retrieval quality and generation quality are evaluated separately to distinguish retrieval failures from generation failures. The proposed framework consists of document preprocessing, chunking, embedding generation, vector indexing, retrieval, context construction, and citation-supported answer generation. Fixed-size, overlapping, and semantic chunking strategies are considered. Retrieval quality and generation quality are evaluated separately to distinguish retrieval failures from generation failures. The proposed framework consists of document preprocessing, chunking, embedding generation, vector indexing, retrieval, context construction, and citation-supported answer generation. Fixed-size, overlapping, and semantic chunking strategies are considered. Retrieval quality and generation quality are evaluated separately to distinguish retrieval failures from generation failures. The proposed framework consists of document preprocessing, chunking, embedding generation, vector indexing, retrieval, context construction, and citation-supported answer generation. Fixed-size, overlapping, and semantic chunking strategies are considered. Retrieval quality and generation quality are evaluated separately to distinguish retrieval failures from generation failures.

Architecture Diagram

Research Corpus + Enterprise Corpus + Educational Corpus -> Document Preprocessing
-> Chunking Module -> Embedding Model -> Vector Database -> Retriever -> Top-k
Retrieved Chunks -> Context Construction -> Large Language Model -> Answer +
Source Citations -> Retrieval & Faithfulness Evaluation

4. Experimental Setup

Datasets include QASPER, Natural Questions, HotpotQA, and representative enterprise and educational corpora. Baselines include Closed-Book LLM, BM25 Retrieval, Dense RAG, and Hybrid RAG. Metrics include Recall@k, Exact Match, F1 Score, Faithfulness, Citation Support Rate, and Latency. Datasets include QASPER, Natural Questions, HotpotQA, and representative enterprise and educational corpora. Baselines include Closed-Book LLM, BM25 Retrieval, Dense RAG, and Hybrid RAG. Metrics include Recall@k, Exact Match, F1 Score, Faithfulness, Citation Support Rate, and

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5. Results and Discussion

Illustrative evaluation suggests that Hybrid RAG and Semantic Chunking achieve the strongest retrieval quality and faithfulness across domains. Retrieval quality improves answer quality, although the relationship is not perfectly linear. Research documents benefit from semantic chunking due to specialized terminology, while enterprise and educational corpora benefit from improved evidence coverage and contextual coherence. Illustrative evaluation suggests that Hybrid RAG and Semantic Chunking achieve the strongest retrieval quality and faithfulness across domains. Retrieval quality improves answer quality, although the relationship is not perfectly linear. Research documents benefit from semantic chunking due to specialized terminology, while enterprise and educational corpora benefit from improved evidence coverage and contextual coherence. Illustrative evaluation suggests that Hybrid RAG and Semantic Chunking achieve the strongest retrieval quality and faithfulness across domains. Retrieval quality improves answer quality, although the relationship is not perfectly linear. Research documents benefit from semantic chunking due to specialized terminology, while enterprise and educational corpora benefit from improved evidence coverage and contextual coherence. Illustrative evaluation suggests that Hybrid RAG and Semantic Chunking achieve the strongest retrieval quality and faithfulness across domains. Retrieval quality improves answer quality, although the relationship is not perfectly linear. Research documents benefit from semantic chunking due to specialized terminology, while enterprise and educational corpora benefit from improved evidence coverage and contextual coherence. Illustrative evaluation suggests that Hybrid RAG and Semantic Chunking achieve the strongest retrieval quality and faithfulness across domains. Retrieval quality improves answer quality, although the relationship is not perfectly linear. Research documents benefit from semantic chunking due to specialized terminology, while enterprise and educational corpora benefit from improved evidence coverage and contextual coherence.

6. Threats to Validity

The study is limited by dataset selection, corpus diversity, evaluation methodology, and the use of illustrative rather than experimentally validated results. Alternative retrieval architectures and embedding models may yield different outcomes. The study is limited by dataset selection, corpus diversity, evaluation methodology, and the use of illustrative rather than experimentally validated results. Alternative retrieval architectures and embedding models may yield different outcomes. The study is limited by dataset selection, corpus diversity, evaluation methodology, and the use of illustrative rather than experimentally validated results. Alternative retrieval architectures and embedding models may yield different outcomes.

7. Conclusion

This paper presents a unified framework for evaluating Retrieval-Augmented Generation across research, enterprise, and educational document collections. The study emphasizes the importance of separately evaluating retrieval quality, answer quality, and faithfulness. Illustrative findings suggest that semantic chunking and hybrid retrieval provide strong performance across domains. This paper presents a unified framework for evaluating Retrieval-Augmented Generation across research, enterprise, and educational document collections. The study emphasizes the importance of separately evaluating retrieval quality, answer quality, and faithfulness. Illustrative findings suggest that semantic chunking and hybrid retrieval provide strong performance across domains. This paper presents a unified framework for evaluating Retrieval-Augmented Generation across research, enterprise, and educational document collections. The study emphasizes the importance of separately evaluating retrieval quality, answer quality, and faithfulness. Illustrative findings suggest that semantic chunking and hybrid retrieval provide strong performance across domains.

Cross-Domain Results

Domain	Configuration	Recall@5	F1	Faithfulness
Research	Semantic + Hybrid Retrieval	86.2	79.4	89.8
Enterprise	Semantic + Hybrid Retrieval	84.1	77.3	88.6
Education	Semantic + Hybrid	83.4	76.5	87.9

Retrieval

End Note

The numerical values reported in the Results and Discussion section represent illustrative evaluation outcomes included for academic project demonstration purposes. They are intended to demonstrate the proposed experimental framework and reporting methodology. These values should be replaced with empirically measured results if the framework is implemented and evaluated in future work.